LANGMUIR FITTING: QUICK REFERENCE CHEAT SHEET

One-page summary of everything for Phase 2

THE LANGMUIR EQUATION (What We're Fitting)

```
q = (qmax × KL × Ce) / (1 + KL × Ce)

q     = Adsorbed amount (mg/g)  ← What we predict
qmax  = Max capacity (mg/g)      ← Parameter (changes with conditions)
KL    = Binding constant (L/mg)  ← Parameter (changes with conditions)
Ce    = Equilibrium conc (mg/L)  ← Calculated from data
```

Key insight: qmax and KL aren't constant—they change with pH, temperature, ions, etc.

THE MODEL WE USE (What's Different)

NOT: Simple 2-parameter Langmuir (ignores factors)
YES: Multi-factor Langmuir with polynomial features

```
Step 1: Take 10 factors (pH, C0, Time, Dose, Temp, Flow, Cl , Hard, CO3, NOM)
Step 2: Add squares (pH², C0², Time², ...)
Step 3: Add interactions (pH×C0, pH×Time, ...)
Result: 66 features from 10 factors

Step 4: Linear regression on 66 features → 67 coefficients
Result: q =   +  ×f  + ... +   ×f
```

Why? Factors change the curve shape → need to capture those changes.

WHAT HAPPENS STEP-BY-STEP

```
[1] Load:          Read CSV (500 samples, 10 factors)
    ↓
[2] Standardize:   Scale to mean=0, std=1 (numbers become comparable)
    ↓
[3] Expand:        10 factors → 66 polynomial features
    ↓
[4] Fit:           Linear regression (minimize prediction error)
    ↓
[5] Predict:       Apply model to get predictions for all 500 points
    ↓
[6] Calculate:     R², RMSE, residuals, diagnostics
    ↓
[7] Visualize:     4 diagnostic plots (save as PNG)
    ↓
```

```
[8] Save:          CSV with predictions, JSON with metadata
```

Runtime: ~1 minute (mostly visualization)

EXPECTED RESULTS AT A GLANCE

Metric	Expected	Interpretation
R^2	0.84-0.87	Explains 84-87% of variance Good!
RMSE	0.9-1.2 mg/g	Average error ±0.9 mg/g (13% of range)
MAE	0.7-0.9 mg/g	Typical error magnitude
Residual mean	0	No systematic bias
Residual std	0.92 mg/g	Residual spread (what ML learns)
Residual dist	Normal	Bell-shaped histogram

Bottom line: $R^2 = 0.85$ is EXCELLENT for a chemical model. Leaves 15% for ML to learn.

THE 4 DIAGNOSTIC PLOTS

```
Panel 1: Actual vs Predicted
   Points follow diagonal line?
   No systematic curve?
  → If YES: Good fit!

Panel 2: Residuals vs Predicted
   Random scatter around zero line?
   Constant width (not funnel)?
  → If YES: Homoscedastic, good!

Panel 3: Residual Distribution
   Bell curve shape?
   Centered at zero?
   No extreme outliers?
  → If YES: Normal distribution, good!

Panel 4: Q-Q Plot
   Points follow straight line?
   Minor deviations at tails OK?
  → If YES: Normal distribution, good!
```

INTERPRETING YOUR RESULTS

R^2 Score

```
What it means: "Model explains X% of variance"
```

```
R² = 0.85 means:
  85% of q variation is explained by Langmuir model
  15% remains as residuals (for ML to learn)

Is 0.85 good?
   YES! For a chemistry model, this is excellent.
   Shows Langmuir is appropriate.
   Leaves good room for ML improvement (15%).

If R² < 0.75?
  ~ Probably OK, but investigate residuals

If R² > 0.95?
   Very good! (But may indicate overfitting in theory)
   Actually fine for Phase 2 (synthetic data is cleaner)
```

RMSE Score

```
What it means: "Average prediction error is X mg/g"

RMSE = 0.92 means:
  Typical prediction off by ±0.92 mg/g
  Range of data is 1.6-8.3 (6.68 total)
  Error is 0.92/6.68 = 13.8% of range

Is 0.92 good?
   YES! <15% of range is good.
   Provides practical confidence intervals.

Rule of thumb:
  < 10% of range = Excellent
  10-20% of range = Good   (You are here)
  20-30% of range = Acceptable
  > 30% of range = Poor
```

Residual Analysis

```
What to look for:

1. Mean   0?
     YES → No systematic bias
     NO → Model over/underestimates

2. Standard deviation   0.92?
     YES → Consistent with RMSE
     NO → High variability (investigate)
```

3. Normal distribution?
    YES → Good for statistics
    NO → Non-normal (usually OK, regression is robust)

4. Random scatter (no patterns)?
    YES → Model captured the pattern
    NO → Model missing something (e.g., non-linearity)

5. No extreme outliers?
    YES → Good data quality
    NO → Investigate unusual conditions

OUTPUT FILES EXPLAINED

langmuir_predictions.csv

Contains: 500 rows, 15 columns

- First 12: Original data (factors + design columns)
- Column 13: q_removal (actual)
- Column 14: q_predicted (model)
- Column 15: residual (actual - predicted)

Use for:

- Phase 3: Analyze residual patterns
- Find worst predictions (largest |residual|)
- Identify where model struggles

langmuir_model_info.json

Contains: Model metadata and metrics

- R^2 = 0.8456
- RMSE = 0.9231
- MAE = 0.7145
- residual_mean = -0.000001
- residual_std = 0.9228

Use for:

- Reference baseline
- Compare with Phase 4 ML results
- Track improvement to hybrid model

langmuir_diagnostics.png

Contains: 4-panel diagnostic plot

- Panel 1: Actual vs Predicted
- Panel 2: Residuals vs Predicted
- Panel 3: Residual Distribution
- Panel 4: Q-Q Plot

```
Use for:
  • Visual validation of fit
  • Check for systematic patterns
  • Identify outliers or issues
  • Report figure for documentation
```

TROUBLESHOOTING QUICK FIXES

Problem	Likely Cause	Quick Fix
$R^2 < 0.70$	Model too simple OR data issue	Check data quality, try degree=3 polynomial
$R^2 > 0.98$	Overfitting (theory)	OK for Phase 2, proceed normally
RMSE > 2.0	Model missing major effects	Check residual plots for patterns
Residuals not normal	Outliers or skewness	Check Q-Q plot, identify outliers
Funnel in residual plot	Heteroscedasticity	OK for Phase 2, ML will learn it
ImportError (sklearn)	Package not installed	`pip install scikit-learn scipy`
File not found	CSV in wrong folder	`cp data/dataset_simulated_500.csv .`

PHASE 2 SUCCESS CRITERIA

Script runs without errors R^2 is 0.80-0.95 (0.84-0.87 expected) RMSE is < 1.5 mg/g (0.9-1.2 expected) Residuals appear normal (bell curve) No systematic patterns in residuals 3 output files created successfully Diagnostic plots look reasonable You understand what was fitted

If all : You're ready for Phase 3!

WHAT COMES NEXT

Phase 3 (Residual Analysis): - Analyze what Langmuir missed - Identify patterns in residuals - Engineer features to capture patterns - Prepare data for ML training

Phase 4 (ML Training): - Train Random Forest, XGBoost, MLP - Predict residuals (not absolute values) - Expected: R^2 0.90-0.93 on residuals

Phase 5 (Hybrid Integration): - Combine: q_hybrid = q_langmuir + q_ml_correction - Achieve: R^2 0.94 (Target!) - Success: 20-35% improvement

---

KEY EQUATIONS TO REMEMBER

Langmuir:

$$q = (q_{max} \times K_L \times C_e) / (1 + K_L \times C_e)$$

R^2 Score:

$$R^2 = 1 - (SS_{residuals} / SS_{total})$$
$$R^2 = 1 - (\Sigma(y - \hat{y})^2 / \Sigma(y - \bar{y})^2)$$

RMSE:

$$RMSE = \sqrt{(\Sigma(y - \hat{y})^2 / n)}$$

Residuals:

```
residuals = y_actual - y_predicted
```

MENTAL MODEL

Think of Phase 2 like this:

```
Question:  "How much fluoride does the carbon remove?"

Chemistry says:
  "Use Langmuir equation with these factors."

We do:
  1. Learn how factors modify Langmuir
  2. Build a model that captures that
  3. Make predictions (should be ~85% accurate)
  4. Calculate what the model missed (residuals)

Result:
  - Chemistry explains 85% (R² = 0.85)
  - Residuals = 15% (opportunities for ML)

Next:
  - ML learns to predict residuals
  - Hybrid model = 85% + 9% = 94% (target!)
```

DECISION TREE: IS MY FIT GOOD?

```
                Is R²  0.80?
                   |
              YES  |  NO
                   |
```

```
  GO TO INVESTIGATE
  PHASE 3  (check data)
       |
Are residuals
 normal?
       |
  YES  |  NO
       |
  GOOD! INVESTIGATE
  PHASE 3  (outliers)
       |
Are there
patterns in
residuals?
       |
  YES  |  NO
       |
EXPECTED  PERFECT!
(Phase 3  RANDOM
 learns)
```

ONE-LINE SUMMARY

Phase 2 Langmuir Fitting: Fit multi-factor Langmuir model to data, achieve R^2 0.85, understand residuals, prepare for ML.

QUICK STATS TABLE

Parameter	Value	Unit
Samples	500	points
Factors	10	original
Features	66	polynomial
Coefficients	67	(+ 66 's)
Sample/Feature ratio	7.6	(good, avoid overfitting)
Expected R^2	0.84-0.87	fraction
Expected RMSE	0.9-1.2	mg/g
Data range	1.6-8.3	mg/g
Error as % range	13.8%	excellent
Runtime	<1	minute
Output files	3	(CSV, JSON, PNG)
Next phase	Phase 3	Residual Analysis

This is your complete Phase 2 reference guide!

Keep this nearby while running Phase 2. Refer to the comprehensive guides for detailed explanations.

Ready? Run: `python phase2_langmuir_fitting.py`

EOF cat /mnt/user-data/outputs/LANGMUIR_CHEAT_SHEET.md